	Marwadi University **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM: Simulate dilation, erosion, opening and closing operation on images.**	
Experiment No: 11	**Date:30-10-2026**	**Enrollment No:92400133121**

Aim: Simulate dilation, erosion, opening and closing operation on images.

Dilation, erosion, opening, and closing are morphological operations used in image processing to manipulate the shape and size of objects in an image.

Dilation:
Dilation expands the boundaries of objects in an image by adding pixels to the object's boundaries. It is achieved by sliding a structuring element over the image and assigning the maximum pixel value within the neighborhood of each pixel.

Erosion:
Erosion shrinks the boundaries of objects in an image by removing pixels from the object's
boundaries. It is achieved by sliding a structuring element over the image and assigning the minimum pixel value within the neighborhood of each pixel.

Opening:
Opening is an erosion operation followed by a dilation operation. It is used to remove noise and small objects from the image while preserving the larger objects' shapes.

Closing:
Closing is a dilation operation followed by an erosion operation. It is used to close small gaps and holes within objects while preserving the overall object shapes.

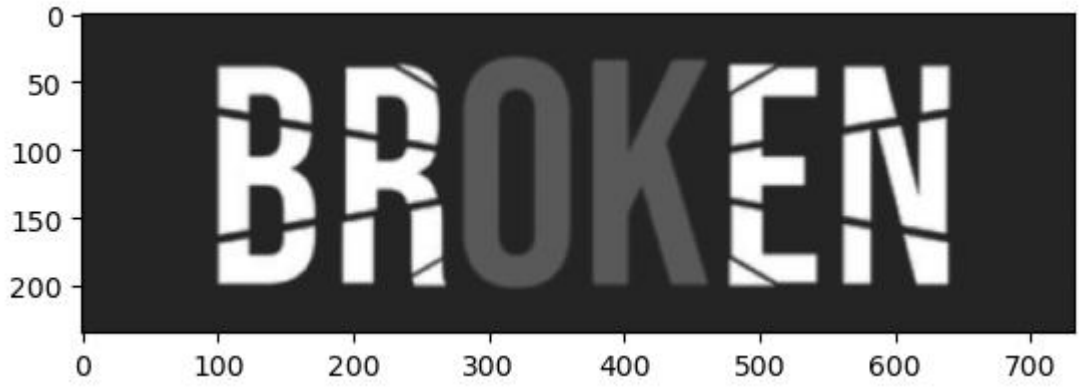
Code:

```
image_path = "images/BROKEN_IMAGE.png"
image = cv2.imread(image_path,0)

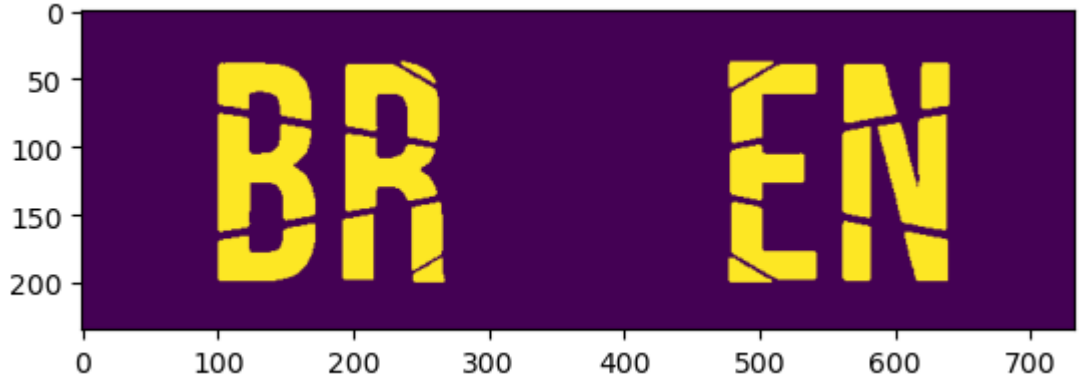

_,binary_image = cv2.threshold(image,123,255,cv2.THRESH_BINARY)

plt.imshow(image,cmap="gray")
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM: Simulate dilation, erosion, opening and closing operation on images.**	
Experiment No: 11	**Date:30-10-2026**	**Enrollment No:92400133121**



plt.imshow(binary_image)



kernel = cv2.getStructuringElement(cv2.MORPH_RECT,(3,3))
print(kernel)

```
[[1 1 1]
 [1 1 1]
 [1 1 1]]
```

→Image←

dilated_image = cv2.dilate(image,kernel,iterations=3)
eroded_image = cv2.erode(image,kernel,iterations=3)
opened_image = cv2.morphologyEx(image, cv2.MORPH_OPEN, kernel)
closed_image = cv2.morphologyEx(image, cv2.MORPH_CLOSE, kernel)

```
plt.figure(figsize=(15,5))

plt.subplot(1,5,1)
plt.title("Original Image")
plt.imshow(image,cmap="gray")

plt.subplot(1,5,2)
plt.title("Dilated Image")
plt.imshow(dilated_image,cmap="gray")

plt.subplot(1,5,3)
plt.title("Eroded Image")
plt.imshow(eroded_image,cmap="gray")

plt.subplot(1,5,4)
plt.title("Opened Image")
plt.imshow(opened_image,cmap="gray")

plt.subplot(1,5,5)
plt.title("Closed Image")
plt.imshow(closed_image,cmap="gray")

plt.show()
```



→BINARY IMAGE←

```
dilated_image = cv2.dilate(binary_image,kernel,iterations=3)
eroded_image = cv2.erode(binary_image,kernel,iterations=3)
opened_image = cv2.morphologyEx(binary_image, cv2.MORPH_OPEN, kernel)
closed_image = cv2.morphologyEx(binary_image, cv2.MORPH_CLOSE, kernel)


plt.figure(figsize=(15,5))

plt.subplot(1,5,1)
```

```
plt.title("Original Image")
plt.imshow(binary_image,cmap="gray")

plt.subplot(1,5,2)
plt.title("Dilated Image")
plt.imshow(dilated_image,cmap="gray")

plt.subplot(1,5,3)
plt.title("Eroded Image")
plt.imshow(eroded_image,cmap="gray")

plt.subplot(1,5,4)
plt.title("Opened Image")
plt.imshow(opened_image,cmap="gray")

plt.subplot(1,5,5)
plt.title("Closed Image")
plt.imshow(closed_image,cmap="gray")

plt.show()
```



Conclusion:

Dilation: The boundaries of objects were expanded.
Erosion: The boundaries of objects were reduced.
Opening: Small noise was removed from the image.
Closing: Small gaps and holes were filled in the image.

→Thus, the morphological operations were studied and their effects on image shapes were observed.